TOPPER'S VAULT 👑

STUDENTS ALERT 🚨

BCA / B.TECH SEMESTER EXAMS

EXPERT PREDICTED IMPORTANT QUESTIONS & SUGGESTIONS

UNIX AND SHELL PROGRAMMING [BCAC602]

(FINAL MASTER EDITION - 200% EXAM READY)

⚡ **EVERY TOPIC UNLOCKED**

⚡ **EXTRA SHELL SCRIPTS ADDED**

⚡ **GUARANTEED EXAM HITTERS**

🎯 STUDY SMART, NOT HARD.

You asked for MORE, and here it is! We have added even more deep-level topics and scripting questions so that you don't miss a single mark.

MODULE 1: Introduction to Unix

1. Explain the Unix System Architecture with a neat block diagram (Hardware, Kernel, Shell, Utilities). 🔥 **V.V. IMP**
2. What are the main differences between Unix and Linux? Name a few popular Linux distributions.
3. Discuss the Unix Directory Structure (Root, /bin, /etc, /dev, /usr, /home) and its hierarchical nature.
4. Write short notes on: (a) POSIX (b) System Call Interface. Explain the role of system calls.

MODULE 2: Unix File Commands

1. Explain basic directory commands: `mkdir`, `rmdir`, `cd`, `ls` (with its various options like -l, -a).

2. What are the different types of files in Unix? (Ordinary files, Directory files, Device/Special files). 🔥 IMP

3. Differentiate between Hard Link and Soft Link. How do you create them? Explain with commands. 🔥 V.V. IMP

4. What is an inode? What information is stored in an inode block? 🔥 HIGH PROBABILITY

5. Explain file permissions in Unix. How do you change file security using absolute (numeric) and symbolic modes using the `chmod` command?

6. Explain file ownership. How do you change the owner and group of a file using `chown` and `chgrp` commands?

7. What is the use of the `touch` command? Explain how to change access time and modification time using `-d` and `-t` options. 🔥 IMP

8. Explain Piping (`|`) and I/O Redirection (>, >>, <). What is the use of the `tee` command?

9. Explain the use of `find` command to locate files based on name, size, and permissions.

10. How do you check the disk space usage of a file system and directory? Explain `df` and `du` commands.

MODULE 3: Text Processing & Manipulation

1. Explain the `sort` command. How do you sort a file numerically, in reverse order, and based on a specific column (using `-k` and `-t` options)? 🔥 IMP

2. Differentiate between `diff` and `cmp` commands used for comparing files.

3. Explain row-wise and column-wise selection from a file using `cut` (with `-d` delimiter and `-f` field options) and `paste` commands.

4. Explain the `wc` command. How do you count the number of lines, words, and characters in a file?

5. What is the use of the `tr` (translate) command? Explain with an example (e.g., converting lowercase to uppercase).

6. What is the use of the `uniq` command? How does it differ from `sort -u`?

7. How do you merge lines horizontally (`join`) and split large files into smaller ones (`split`)?

MODULE 4: Utility Commands

1. Write the syntax and use of the following utility commands: `cal`, `date`, `who`, `echo`.
2. How is the `bc` command used for mathematical calculations in Unix? Explain `pr` command for formatting files for printing.
3. How do you compress and archive files in Unix? Differentiate between `tar`, `gzip`, `zip`, and `unzip` commands. 🔥 **IMP**

MODULE 5: Vi Editor

1. Explain the three basic modes of operations in the Vi editor (Command mode, Insert mode, Last-line/Ex mode). How do you switch between them? 🔥 **V.V. IMP**
2. Differentiate between insertion commands: `i`, `I`, `a`, `A`, `o`, `O` in Vi editor.
3. Write the Vi editor commands for:
    - Deleting a line, deleting a word, copying a line, pasting text. (dd, dw, yy, p)
    - Navigating using arrow keys alternatives (h, j, k, l).
    - Undo (u) and Redo (`Ctrl+r`) operations.
    - Saving and quitting (`:wq`, `ZZ`, `:q!`).
4. How do you perform Global Search and Replace in a file using the Vi editor? Explain the syntax `:s/old/new/g`. 🔥 **IMP**

MODULE 6: Grep and Awk Commands

1. What is the `grep` command? What are Regular Expressions? Explain the use of meta-characters (^, $, *, .) with grep. 🔥 **V.V. IMP**
2. Explain the usage of grep with options like `-i` (ignore case), `-v` (invert match), `-c` (count), and `-n` (line number).
3. Explain the architecture of an awk program. What is the significance of `BEGIN` and `END` sections? 🔥 **HIGH PROBABILITY**
4. Explain the built-in variables in AWK: `FS`, `OFS`, `NR`, and `NF`.

5. How do you use `if-else` statements and `loops` in an AWK script? Explain string functions in AWK (length, substr, index).

MODULE 7: Process Management in Unix

1. What is a Process? Draw and explain the Unix Process State Transition Diagram. 🔥 **V.V. IMP**

2. What is a Zombie Process and an Orphan Process? How are they created? 🔥 **V.V. IMP**

3. Explain the `init` process and the Unix login process in detail.

4. Explain the purpose of the following system calls: `fork()`, `wait()`, `getpid()`, `getppid()`. What does `fork()` return to the parent and child processes?

5. What is Inter-Process Communication (IPC)? Explain how the `pipe()` system call and messages are used for IPC. 🔥 **IMP**

6. Discuss the `ps`, `top`, `vmstat`, and `nice` commands. Differentiate between `top` and `vmstat` (swapped memory vs real time). How does the `nice` value affect process scheduling?

MODULE 8: Shell Programming

1. What is a Shell? Differentiate between Bourne Shell (sh), C Shell (csh), and Korn Shell (ksh). 🔥 **IMP**

2. Explain the difference between Single Quotes (`' '`), Double Quotes (`" "`), and Backticks/Backquotes (`` ` ` ``) in shell scripting. What is Command Substitution?

3. What is the significance of the backslash (`\`) escape character in shell?

4. What are Command Line Parameters (Positional Parameters)? Explain the significance of `$0`, `$1`, `$#`, `$*`, `$@`, and the exit status `$?`. 🔥 **HIGH PROBABILITY**

5. How do you take user input using the `read` command? How do you perform arithmetic evaluation using the `let` command?

6. Explain the `case...esac` statement in shell scripting with a suitable example. 🔥 **IMP**

7. Explain the use of `if-elif-else` statement and looping constructs (`for`, `while`, `until`) in shell scripts with syntax.

⭐⭐ TOPPER'S PREDICTED MEGA QUESTIONS ⭐⭐

(Guaranteed questions compiled from past year trend analysis + Practical Labs)

Group A (Short Concepts - 2/3 Marks)

- What is the function of the Kernel in Unix?
- What does the `tee` command do?
- Difference between `cmp` and `diff`.
- What is an inode?
- What is the significance of the `/etc` directory?
- How do you change the permissions of a file to rwxr-xr-- using absolute mode?
- What are Zombie and Orphan processes?
- What is an exit status ($?) and why is it useful?
- What is the difference between `while` and `until` loop in shell?
- What does the command `chmod 755 filename` mean?
- What is the difference between du and `df` commands?

Group B (Descriptive & Analytical - 5/7 Marks)

- Draw and explain the Unix Architecture.
- Differentiate between Hard link and Soft link with proper commands.
- Explain file ownership and permissions in Unix. (chmod, chown)
- Explain the Process State diagram in Unix.
- Discuss the role of IPC (Inter-Process Communication) in Unix.
- Explain the `fork()` system call. What happens when a process calls fork()?
- What are Regular Expressions? Explain meta characters used with grep.
- Explain the `BEGIN` and `END` blocks in AWK with an example.
- Explain the three modes of the Vi editor and how to switch between them.
- Explain positional parameters ($#, $*, $?, $1) used in shell scripts.
- Explain the different ways to insert text in Vi editor (i, I, a, A, o, O).

Group C (Must-Do Shell Scripts & Lab Scenarios - 10 Marks)

- 1. Write a shell script to create a Basic Calculator using the `case` statement.
- 2. Write a shell script to generate the Fibonacci series up to N terms.

- 3. Write a shell script to reverse a given number.
- 4. Write a shell script to check whether a given number is Prime or not.
- 5. Write a shell script to find the Factorial of a given number.
- 6. Write a shell script to find the greatest of three numbers supplied as command line arguments.
- 7. Write a shell script to check whether a string is a Palindrome or not.
- 8. Write an AWK script to print employee details whose salary is greater than 10000 from a file "emp.txt".
- 9. Write a command pipeline to display the top 10 largest files in a directory using du and `sort`.
- 10. Explain the use of sed command to find and replace a specific word in a file, and how to delete a specific line.

```
# Basic Calculator using case (Highly Repeated)
echo "Enter two numbers:"
read a b
echo "Enter Choice: 1.Add 2.Sub 3.Mul 4.Div"
read ch
case $ch in
  1) res=`expr $a + $b` ;;
  2) res=`expr $a - $b` ;;
  3) res=`expr $a \* $b` ;;
  4) res=`expr $a / $b` ;;
esac
echo "Result : $res"
```